STA 445 S26 Assignment 5

2026-03-19

##Problem 1 Convert the following to date or date/time objects.

a. September 13, 2010.

```
mdy("September 13, 2010")
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```
mdy("Sept 13, 2010")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

The month abbreviation for part b is incorrect.

c. Sep 13, 2010.

```
mdy("Sep 13, 2010")
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs.

```
mdy("S 13, 2010")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

The month abbreviation must be either the full name or three digits it appears.

e. 07-Dec-1941.

```r
dmy("07-Dec-1941")
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

```r
dmy("1-5-1998")
```

```
## [1] "1998-05-01"
```

I could be wrong because I'm not sure if we're talking about January 5th or May 1st, 1998. There's no real way of intending whether the intention is mdy or dmy.

g. 21-5-1998. Comment on why you know you are correct.

```r
dmy("21-5-1998")
```

```
## [1] "1998-05-21"
```

I know that this one is dmy because there are not 21 months, so it could not be mdy.

h. 2020-May-5 10:30 am

```r
ymd_hm("2020-May-5 10:30 am")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
ymd_hm("2020-May-5 10:30 am", tz = "US/Pacific")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
ymd_hm("2020-May-5 10:30 am", tz = "America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

a. Calculate the date of your 64th birthday.

```r
dob <- mdy("April 9, 2003")
dob + years(64)
```

```
## [1] "2067-04-09"
```

b. Calculate your current age (in years).

```r
(age <- interval(dob, today()) %>%
  as.period() %>%
  year())
```

```
## [1] 22
```

c. Using your result in part (b), calculate the date of your next birthday.

```r
(next_bday <- dob+years(1+age))
```

```
## [1] "2026-04-09"
```

d. The number of *days* until your next birthday.

```r
as.period(interval(today(), next_bday), unit = "days")
```

```
## [1] "21d 0H 0M 0S"
```

e. The number of *months* and *days* until your next birthday.

```r
as.period(interval(today(), next_bday))
```

```
## [1] "21d 0H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
mdy_h("May 8, 2015 3 pm", tz = "US/Arizona") %>%
  with_tz(tz = "NZ")
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

Problem 4

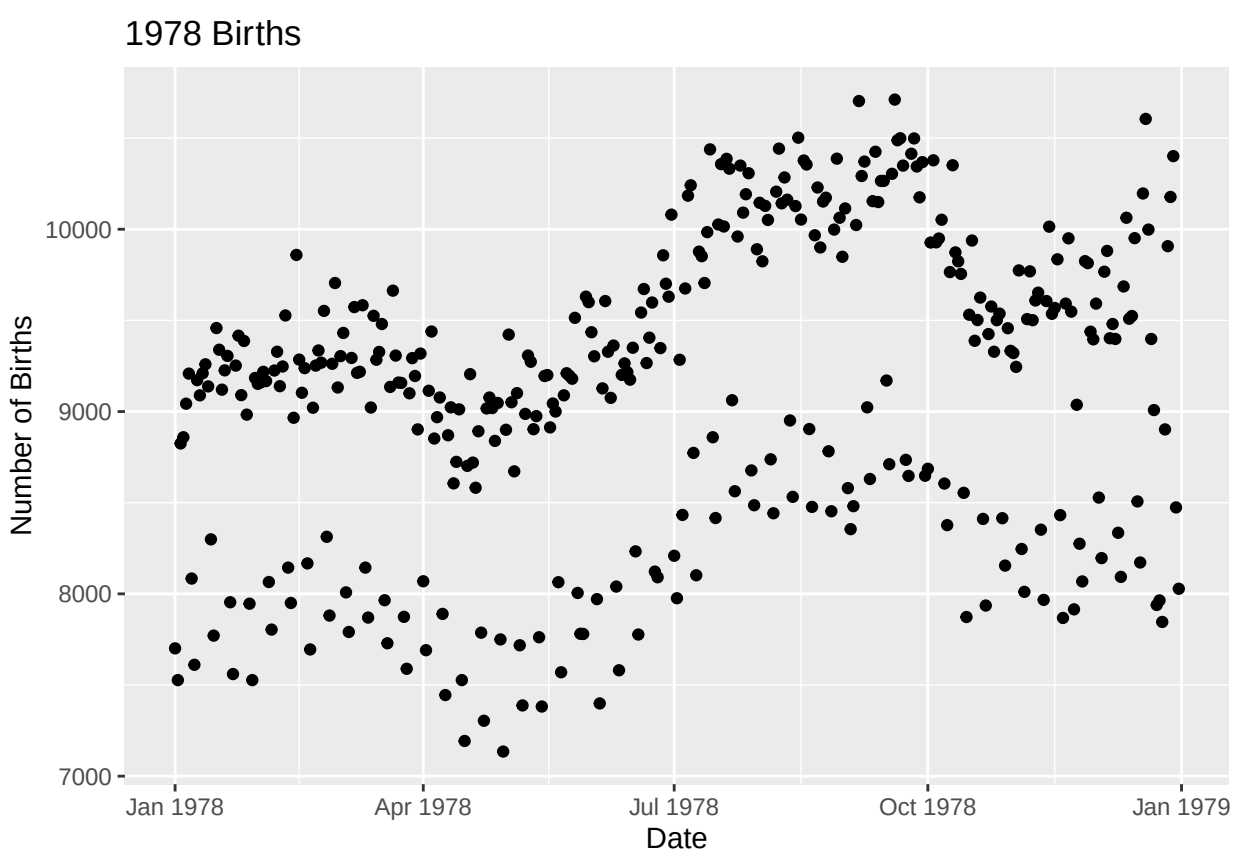
It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```
Births78 <- select(Births78, date, births)
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```
ggplot(Births78, aes(x = date, y = births)) +
  geom_point() +
  labs(
    title = "1978 Births",
    x = "Date",
    y = "Number of Births"
  )
```

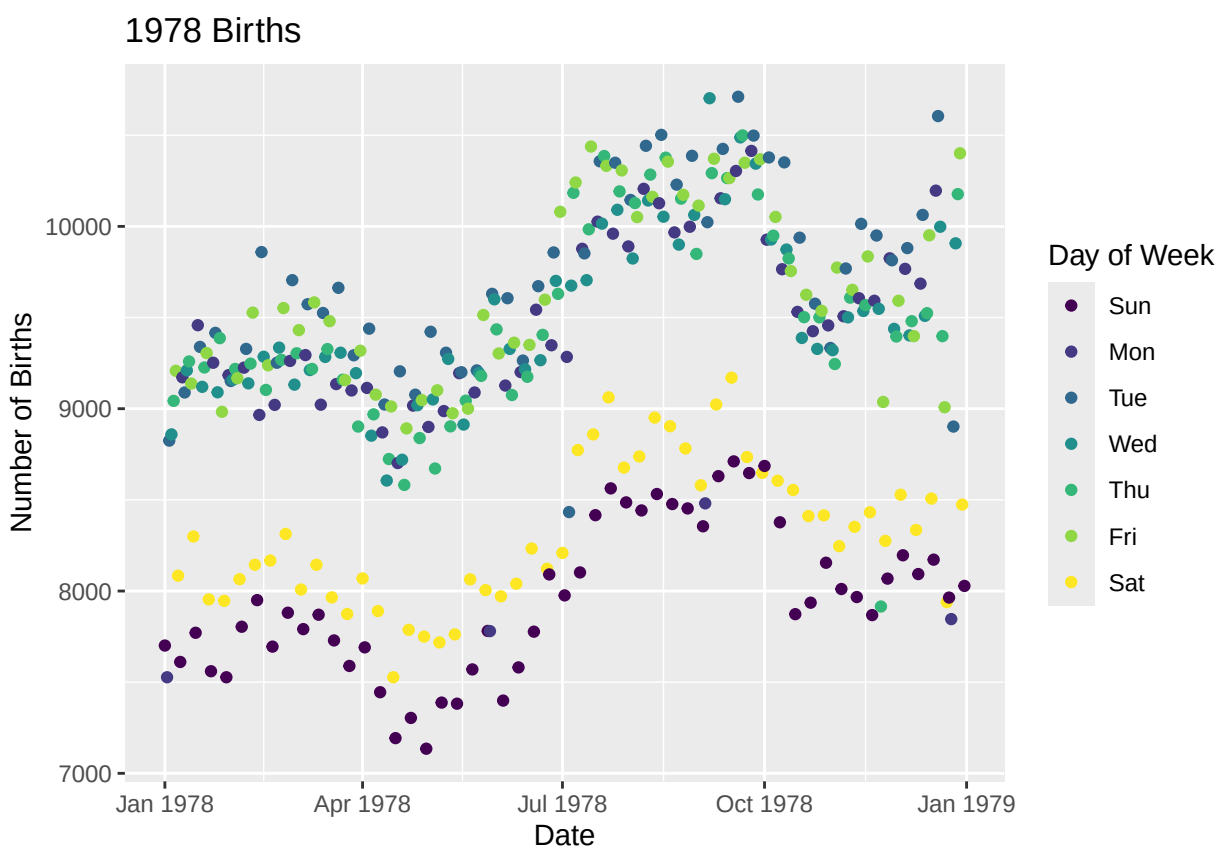


The number of births does appear to oscillate rather noticeably on a week-to-week basis. I think that this is just do to hospital availability more than anything else. There are also more births into the

summer/early fall, and I think that this is because nine months previous to that, the holidays are around and the weather outside is less enjoyable, so people spend more time together indoors.

c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
Births1978 <- Births78 %>%
  mutate(dow = wday(date, label = TRUE)) %>%
  select(dow, date, births)
```

d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(Births1978, aes(x = date, y = births, color = dow)) +
  geom_point() +
  labs(
    title = "1978 Births",
    x = "Date",
    y = "Number of Births",
    color = "Day of Week"
  )
```